Blockchain: Developing Smart Contracts Using Standards and Open Source Libraries

Though the blockchain is very popular these days, there is a significant gap in the way it works and how most of us understand it. Most of its resources such as smart contracts are still a work in progress. Let's talk a bit more about them.

Bitcoin and Ethereum are the two most popular cryptocurrencies today, with the latter being more developer-friendly. A program that runs on the Ethereum blockchain as a collection of code (its functions or rules) and data (its state) that resides at a specific address on this blockchain is a smart contract. A developer would need to learn how to code in smart contract language and have enough ETH (Ethereum currency) to pay as Gas fees (call it cost of doing business or initiating a transaction) for deployment of the contract.

In simpler terms, a smart contract can be considered as an open API that can call other APIs. A chain of smart contracts can make a serious business transaction. In the long run, the number of libraries is going to increase and development of smart contracts is going to be much easier and more efficient, targeted at replacing highly skilled professions such as lawyers or government contract agencies that focus on governance through compliance.

Development environments

Blockchain is an ever-evolving field, and there is no definitive approach or set pattern for the development of smart contracts. However, there are a few open source tools that have proved to be effective as development environments like Truffle, Embark and Builder. These tools efficiently handle compilation, deployment, debugging and upgradation of contracts with the sophistication of running unit tests.

Development of smart contracts requires a set of tools, most of which are available in open source. With trial-and-error, the appropriate environment for the development of smart contracts can be established. The good news is that there is quite a bit of information available on the Web for how to go about doing this. Also, the technology has been in the works for more than five years, bringing in some stability in the tools with enthusiastic developer forums putting in troubleshooting efforts.

Standards

The Ethereum community has defined multiple standards in the form of ERCs (Ethereum requests for comments). This is a document used by smart contract programmers to refer to the rules Ethereum based tokens must comply with. There is a thriving community of developers, enthusiasts and organisations across the globe working to standardise the use of Ethereum. Requests for Ethereum improvement proposal (EIP) are maintained on GitHub. At the time of writing this article, there are nearly 5000 requests suggesting the increased adoption and